Characterization and optimization of Skipper CCDs for UV astronomy

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The Network for Ultraviolet Astronomy - Laboratoire d'Astrophysique de Marseille - 6-10 October, 2025

Astronomy & Steward Observatory

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Introduction

- Observation of faint astronomical objects in the **low-signal, low-background photon-counting regime** with CCDs/CMOS are limited by the readout noise (RN).

$$\sigma_{\text{obs}} = \sqrt{\sigma^2_{\text{shot}} + \sigma^2_{\text{read}}}$$

- Poses challenges for next-generation experiments focusing on:
  - Multi-object spectroscopy of faint stars and galaxies
  - High-cadence searches for short-duration transients
  - Direct exoplanet imaging requires resolving single electrons with high-dynamic range in order to achieve high S/N and contrast on dim exoplanets
  - UV circumgalactic medium (CGM) studies mainly rely absorption ⇒ **next-generation of CGM studies will attempt to map the emission from the diffuse CGM = photon-starved regime**

Skipper CCDs, with low RN and high UV quantum efficiency (QE) are prime detector candidates for these next generation studies.

What are Skipper CCDs?

- Benefits from advantages of standard CCDs (stability, linearity, dynamic range, quantum efficiency, low dark-current) with the unique ability to perform **multiple non-destructive readouts**:
  - Averaging N_{samp} readouts of the same pixel ⇒ readout noise scaled as $\sigma_N \propto 1/\sqrt{N_{samp}}$
  - … at the cost of increased readout times $t \propto 1/RN^2$
  - Readout sequence can optimized for specific science cases to reach the desired S/N in the least amount of time.
  - N_{samp} can be configured on a per-pixel basis = "smart readout" or ROI selection
  - + increase the number of amplifiers (4 up to 128; see R. Smith NSF Award Number 2308380)

$$\mathrm{S/N} = \frac{\mathrm{S_{src}t_{exp}}}{\sqrt{(\mathrm{S_{src}} + \mathrm{S_{bkg}} + \mathrm{S_{dark}})\mathrm{t_{exp}} + \mathrm{N_{samp}}\sigma^2_{\mathrm{read}}}}$$

$$\mathrm{t_{read}} = \mathrm{N}_{row}[((\mathrm{t_{pixel}N}_{samp} + \mathrm{t_H})\mathrm{N}_{col} + \mathrm{t_V}]$$

What are Skipper CCDs?

Source: Tiffenberg et al. 2017, arXiv:1706.00028

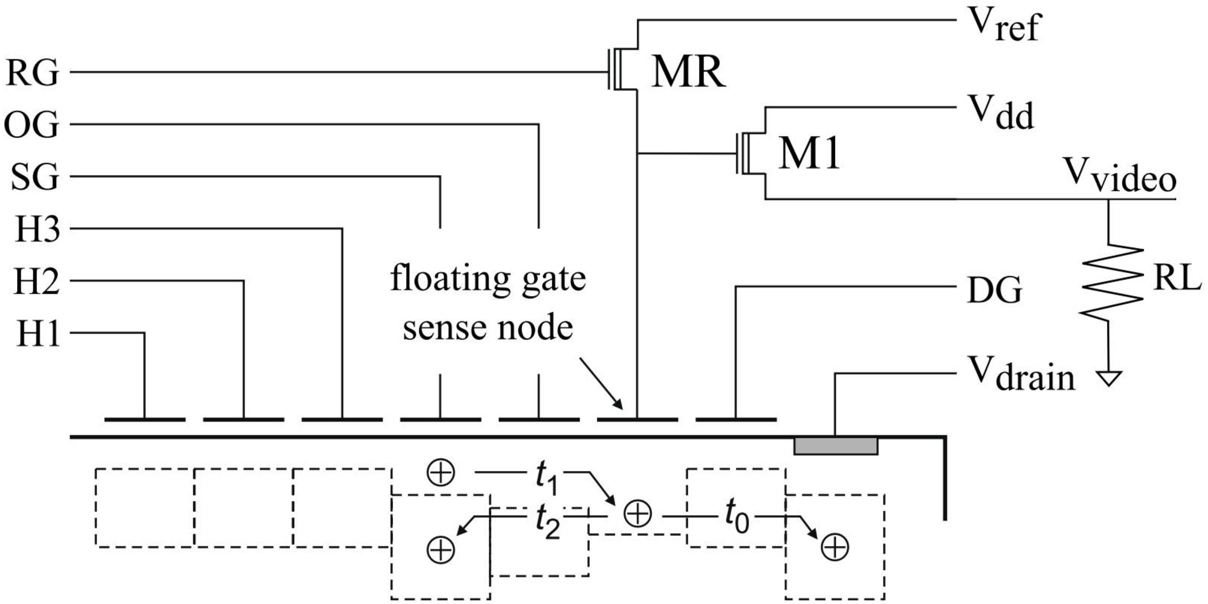


Figure 1 – Diagram of a Skipper CCD readout stage. The floating gate allows non-destructive measurement of the charge at the sense node.

What are Skipper CCDs?

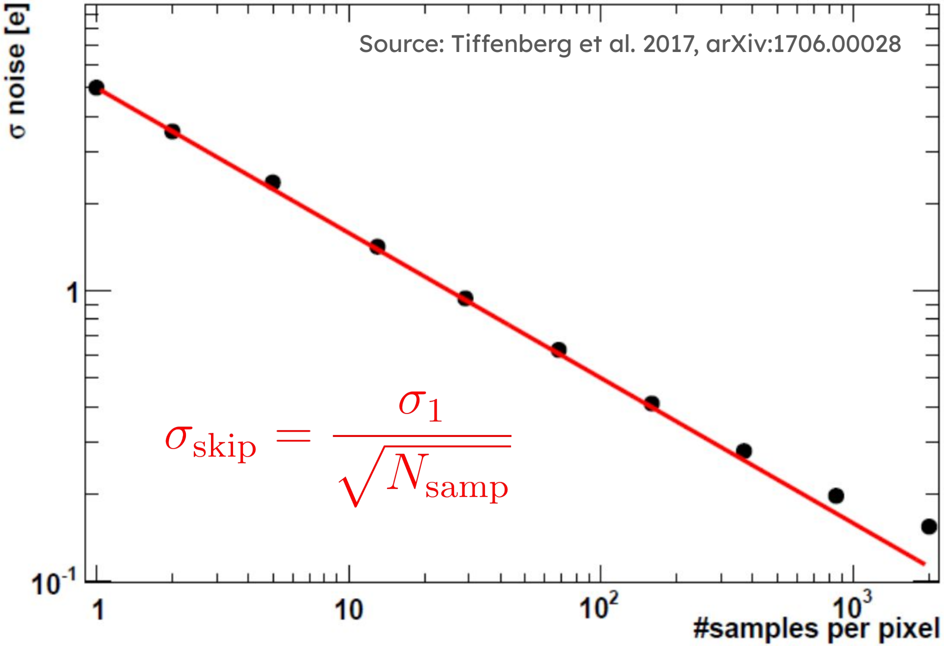


Figure 2 – Measured readout noise as a function of the number of non-destructive readout samples per pixel for the Skipper CCD.

What are Skipper CCDs?

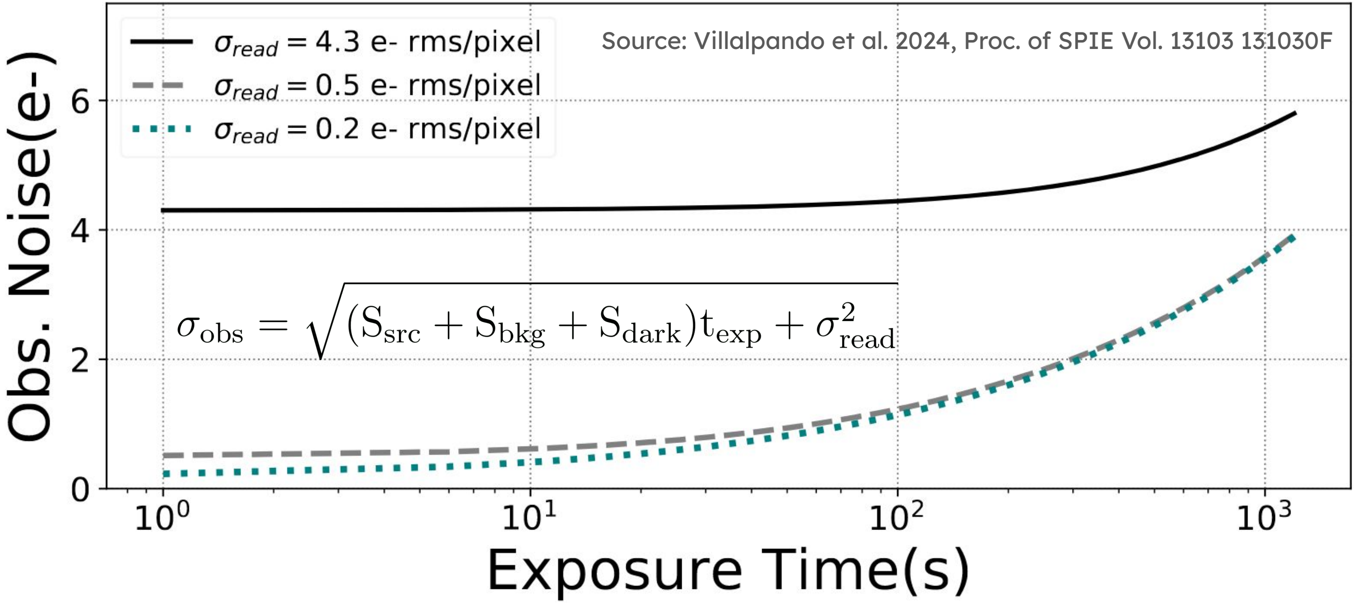


Figure 3 – Interplay between between exposure time (count level) and the readout noise. The total observation noise as a function of the exposure time. The total observation noise includes the background shot noise. For short exposure times, the shot noise is subdominant to the readout noise.

Previous use cases of Skipper CCDs

- Dark matter (DM) direct detection experiments:
  - Theoretically well-motivated sub-GeV DM models = light dark matter
  - DM particles interacting with electrons
  - Requires high-sensitivity + low-noise (dark current + readout) technology + **readout time is not critical** ⇒ Skipper CCDs
  - SENSEI, OSCURA, DAMIC

Source: Brenda A. Cervantes-Vergara et al. 2023
JINST 18 P08016

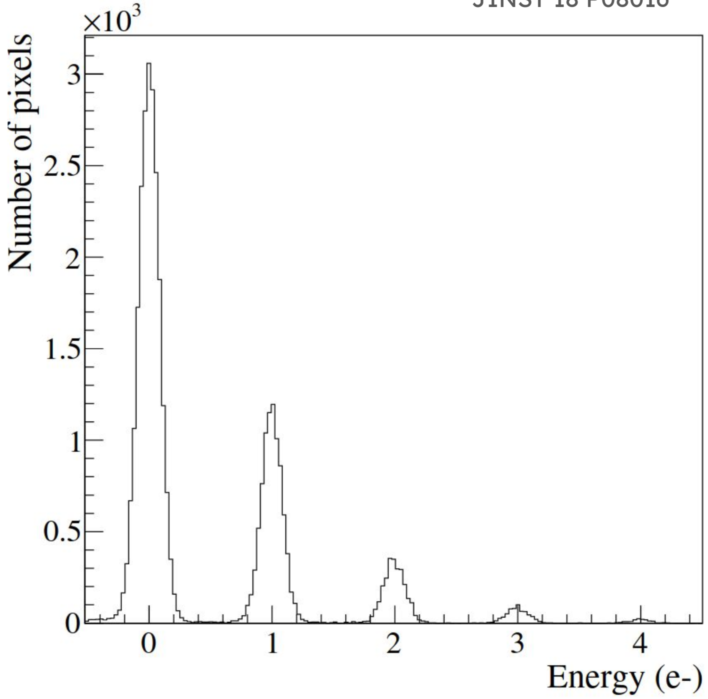
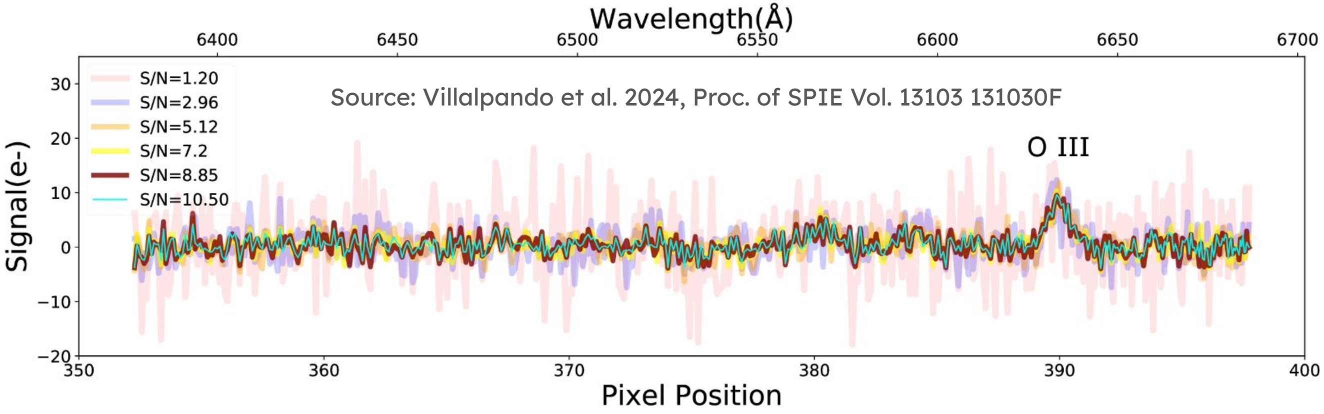


Figure 4 – Charge pixel distribution from acquisition with N_{samp} = 1225 samples per pixel, illustrating the electron-counting capability of the first fabricated Oscura prototype skipper-CCDs.

Previous use cases of Skipper CCDs

- First astronomical application for the SOAR Integral Field Spectrograph (Villalpando et al. 2024, PASP, 136:045001)



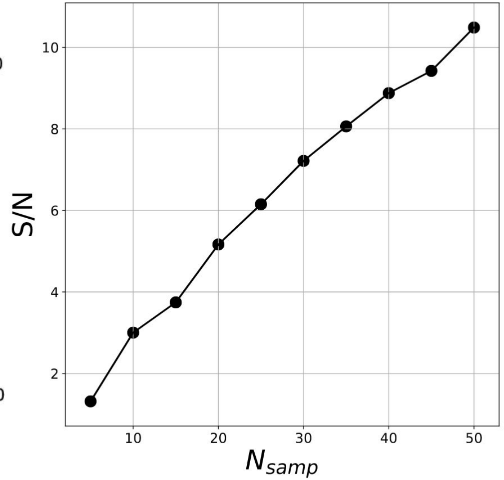


Figure 5 – Left: Spectrum from a t_{exp} = 600 s SIFS-Skipper CCD observation of an ELG identified in the DESI Early Data Release. The OIII emission line is detected at λ ~ 6628.8 Å with different S/N ratios depending on the achieved readout noise (colored lines). **Right:** S/N as a function of N_{samp}. **In the readout-noise-dominated regime, it is possible to increase the SIFS signal-to-noise from ~1.2 to ~10.5 by achieving sub-electron readout noise levels.**

Applications to CGM science

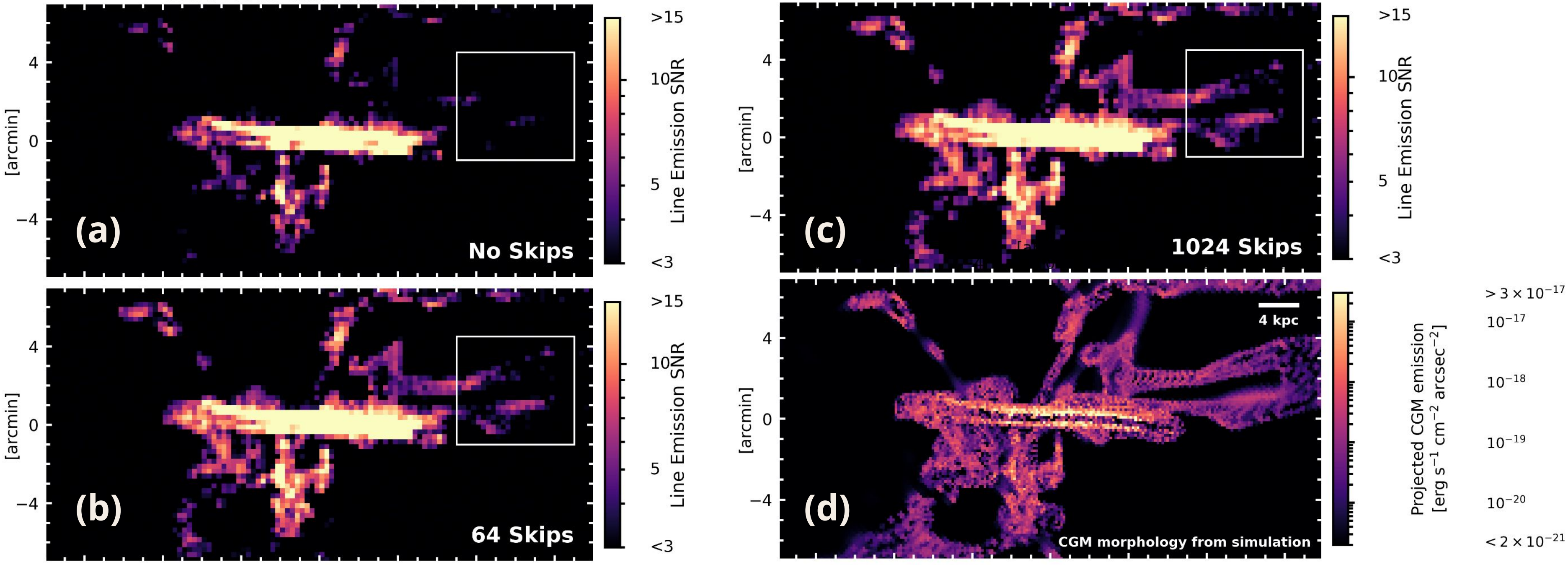
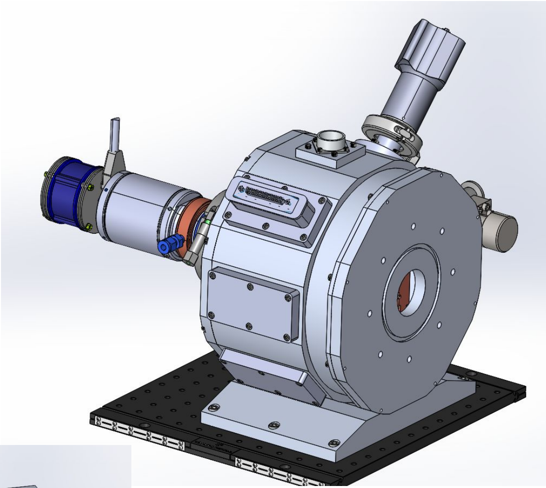


Figure 6 – Panels (a), (b) and (c) Improvement of S/N with increasing N_{samp} (1, 64 and 1024) for constant exposure time. **Panel (d)** Original emission map from simulation in erg/s/cm^2/arcsec^2.

Vacuum dewar setup

- Development of a test bench for in-lab characterization:
  - Hot-swappable onto a vacuum UV dewar for PTC and QE measurements.
  - Temperature control:
    - Cryotel GT Cryocooler
    - Lakeshore 350 controller
    - Heat resistors on the CCD backplate
  - Vacuum:
    - Pfeiffer HighCube Neo 80 pump w/ PKR 251 gauge
    - MKS 971B gauge
  - Detector controllers:
    - Low-Threshold Acquisition Controller (LTA)
    - STA Archon controller



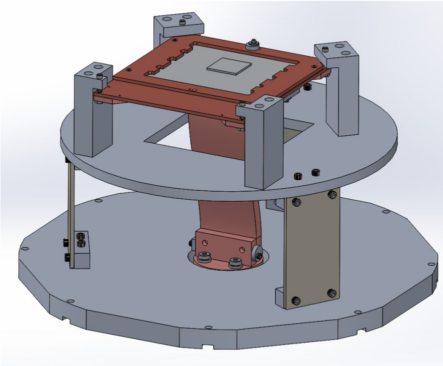


Figure 7 – CAD overview of the Skipper CCD vacuum dewar test setup

MUV-VIS characterization facility

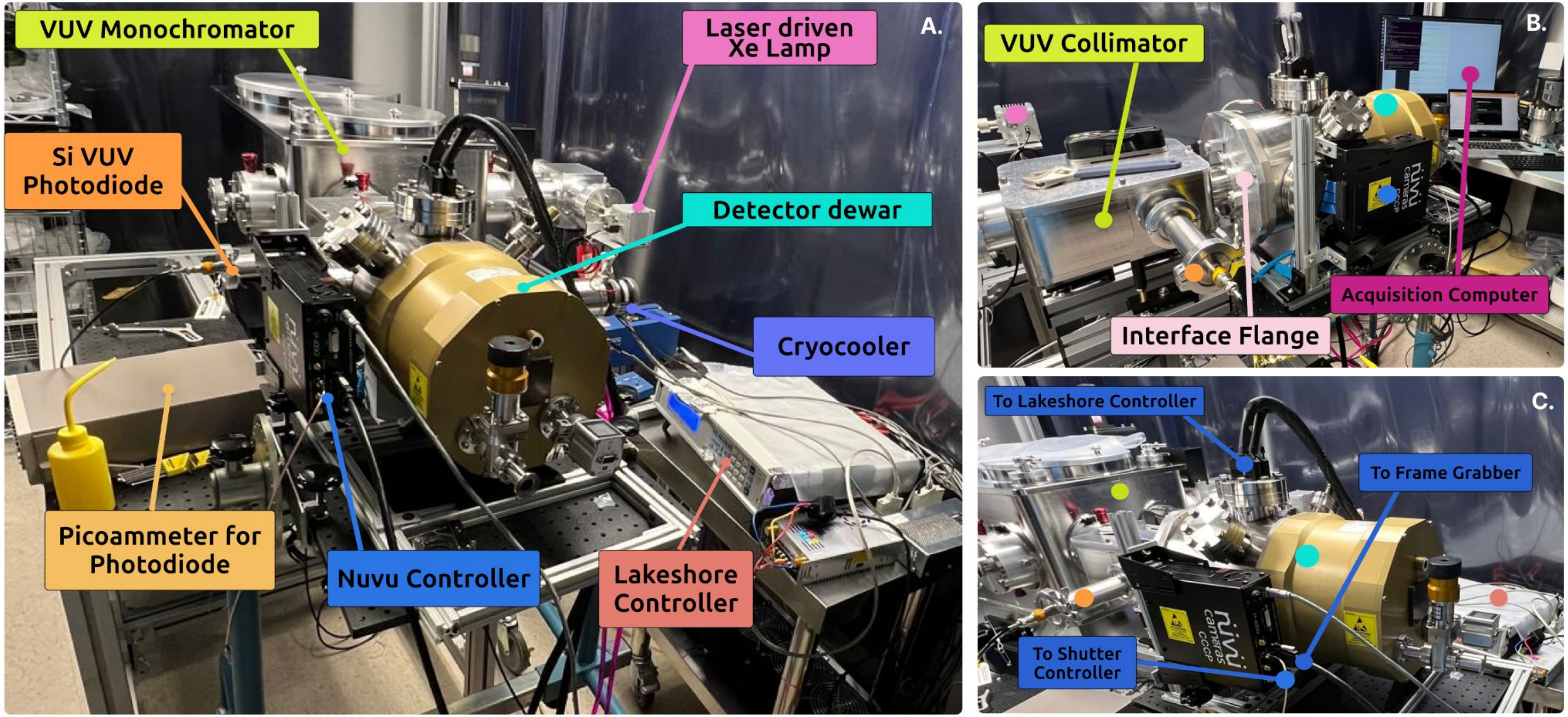


Figure 8 – Clean room ultra-high vacuum QE and PTC measurements bench setup for the EMCCD CCD201-20 from Teledyne-e2v

Past results using the MUV-VIS facility

- Dark-current investigations of EMCCD CCD201-20 from Teledyne-e2v (A. Khan et al., JATIS, Vol. 11, Issue 4, 042204, June 2025)
- Princeton Instruments Sophia XO-2048B for Super-LOTIS

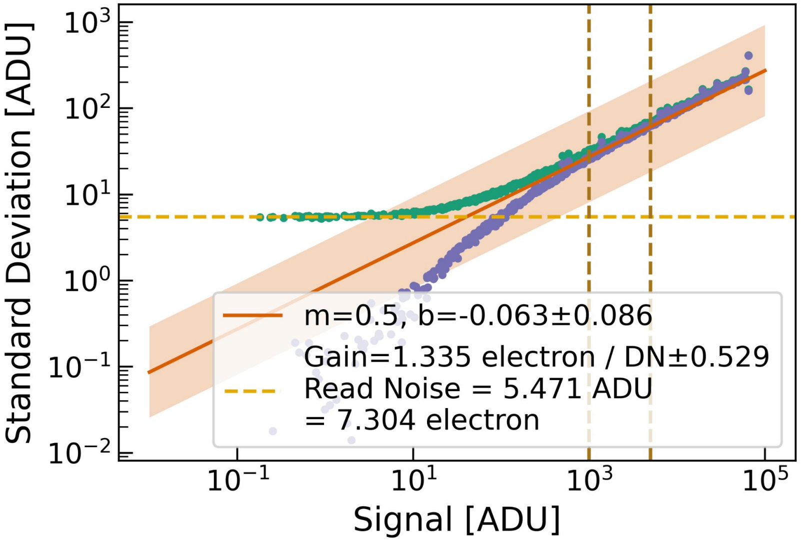


Figure 9 – Photon transfer curve for the Princeton Instruments Sophia XO-2048B camera of Super-LOTIS

Readout noise analytical predictions

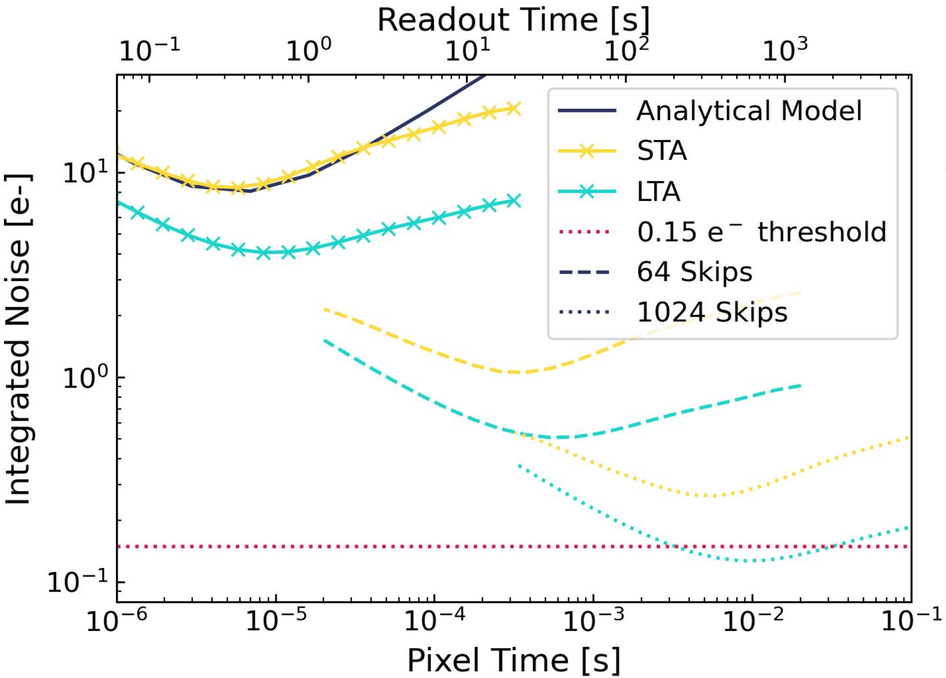
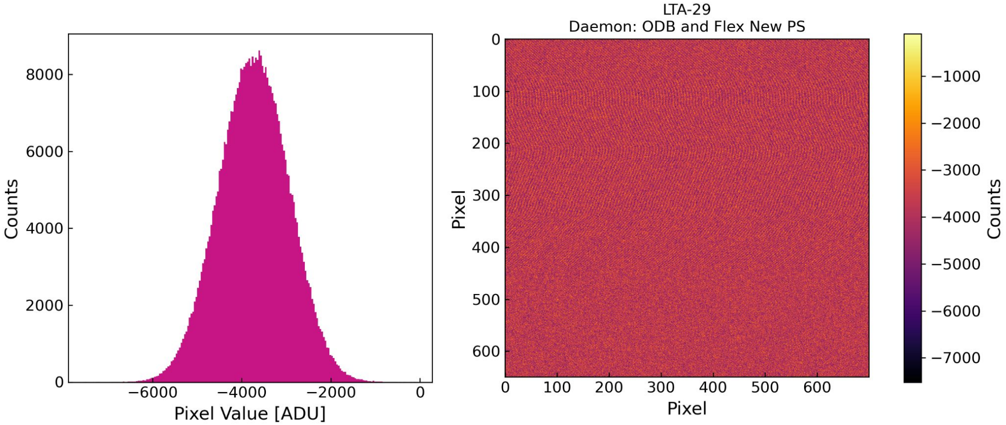


Figure 10 – Comparison of RN predictions for STA (solid) and LTA (dashed) controllers with N_{samp} = [1, 64, 1024].
The target threshold of 0.15 e^- is reachable at the cost of ~ 600 seconds readout time.

Current status and future plans

- Completing manufacturing and assembly of the Skipper CCD dewar
- Testing the setup with OSCURA detectors and compare results to the original experiment
- Thinning of silicon wafers at JPL and application of UV-enhanced AR coatings and backside passivation (delta-doping)
- In-house detectors packaging at the Imaging Technology Laboratory (ITL)
- Upgrading the vacuum monochromator for high-precision QE measurements
- In-lab detectors comparison and characterization studies (read noise, dark current, PTC, QE)

Figure 11 – Latest bias image and histogram of an Oscura Skipper CCD using the LTA controller.



Summary

- In exposures with few counts (photon-starved regime), reducing the readout noise can lead to significant gains in signal-to-noise.
- Skipper CCDs benefits from well-established advantages of conventional astronomical CCDs (i.e., stability, linearity, dynamic range, quantum efficiency, and radiation response) with the capability of performing multiple non-destructive pixel reads to reach ultra-low noise (< 0.1 e rms/pixel).
- S/N can be dynamically adjusted on a exposure-by-exposure or a pixel-by-pixel basis.
- Skipper CCDs offer a promising solution for space-based UV CGM studies.